

API Security Risk Analysis Report

Target API Under Test: ReqRes API Engine (<https://reqres.in>)

Testing Tools Utilized: Postman Web Client

Test Scenario	Executed Action	Observed Result	Security Finding
Authentication Control	Added the missing x-api-key header with the generated token into the Postman Headers tab.	The response instantly shifted from a 401 Unauthorized error to a successful 200 OK status, printing out the JSON user directory profile strings cleanly.	Passed. The endpoint securely prevents anonymous threat actors from scraping user account data records without passing a valid authorization key first.
Rate Limiting Check	Triggered the blue Send button 5 to 6 times rapidly back-to-back to simulate automated API flooding.	The API returned a 200 OK status response for every single click without throttling or blocking the request.	Vulnerable / Risk Identified. The absence of a standard rate-limiting threshold leaves the endpoint open to potential automated resource depletion loops or credential brute-forcing.